



Bangladesh University of Engineering & Technology

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Energy Conversion II Laboratory**

**Final Project Report
Section : C2 Group:02**

**Rewinding of Three Phase Induction Motor For Water Pumping
Applications**

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List of Symbols/Nomenclature

Ns	Synchronous speed of the rotating magnetic field	RPM
N (or Nr)	Actual rotor mechanical speed	RPM
f	Supply frequency	Hz
P	Number of magnetic poles	—
s	Slip	— (or %)
q	Slots per pole per phase	—
γ	Electrical angle between adjacent slots	degrees (°)
Vline	Applied line-to-line voltage	V
Vph	Per-phase stator voltage	V
Iline	Line current	A
Iph	Phase current	A
R1	Stator winding resistance	Ω
R2'	Rotor resistance referred to the stator side	Ω

R_c	Core-loss resistance	Ω
X_1	Stator leakage reactance	Ω
X_2'	Rotor leakage reactance referred to the stator side	Ω
X_m	Magnetizing reactance	Ω
Z_{br}	Blocked-rotor impedance	Ω
ϕ	Impedance / power factor angle	degrees (°)
V_{th}	Thevenin equivalent voltage	V
R_{th}	Thevenin equivalent resistance	Ω
X_{th}	Thevenin equivalent reactance	Ω
ω_{sync}	Synchronous angular velocity	rad/s
T_{dev}	Developed electromagnetic torque	N·m
T_{start}	Starting (locked-rotor) torque	N·m
T_{max}	Maximum (breakdown/pull-out) torque	N·m
T_{FL}	Full-load torque	N·m
Q	Volumetric water discharge flow rate	L/s (or m ³ /h)

Rewinding of a Three Phase Induction Motor For Water Pump Applications

Abstract: This project presents a comprehensive, end-to-end engineering workflow encompassing the physical reconstruction, parameter characterization, mathematical modeling, and experimental performance evaluation of a three-phase squirrel-cage induction motor (0.37kW, 400V, 1.1A, 50Hz, 4-pole) for agricultural and domestic water pumping applications. While it is calculated for 60 Hz instead of 50 Hz for calculation part and matlab simulation part. The methodology began with a complete stator teardown, physical dimension inspection (6.5cm inner diameter, 13cm outer diameter, and 53mm slot length), and a precise 24-slot copper rewinding process utilizing 24SWG super-enamelled copper wire with 99 turns per slot in a Y-connected configuration. Following physical restoration, standard electrical testing—including DC resistance measurements, no-load tests, and blocked-rotor tests—was systematically performed to extract critical equivalent-circuit parameters, yielding a stator resistance (R_1) of 28.1Ω , referred rotor resistance (R_2') of 18.23Ω , core-loss resistance (R_c) of 654.75Ω , leakage reactances ($X_1 = X_2$) of 68.87Ω , and a magnetizing reactance (X_m) of 121.9Ω . Utilizing these experimentally derived values, a Thevenin-based mathematical model was implemented in MATLAB to simulate the electromechanical torque-speed characteristics. The simulation successfully established key operational thresholds, including a starting torque (T_{start}) of 0.4 Nm, a breakdown pull-out torque (T_{max}) of 1.3 Nm at 1515RPM, and a full-load operating point of 0.8 Nm at 1710RPM. Finally, coupling the motor mechanics with centrifugal pump equations validated real-world fluid dynamics, achieving a volumetric discharge rate of 12.00L/s ($43.2m^3/h$) at rated speed. This work successfully bridges practical electrical machine repair with rigorous theoretical modeling, ensuring enhanced reliability, efficiency, and predictability for deployed pumping systems.

Chapter 1: Introduction

1.1 Project Overview

The project titled “Reconstruction of Three-Phase Induction Motor for Water Pumping Application” focuses on the reconstruction, analysis, and practical application of a three-phase squirrel-cage induction motor for a water pumping system. Three-phase induction motors are widely used in industrial and agricultural pumping applications because of their simple construction, high reliability, ruggedness, and suitability for continuous operation. In this project, the motor is reconstructed through motor disassembly and stator restoration, followed by copper rewinding, while the existing squirrel-cage rotor is retained.

1.2 Why Squirrel-Cage Induction Motor?

A squirrel-cage induction motor is particularly suitable for water pumping applications due to its robust and simple rotor construction. Unlike a wound-rotor motor, it does not require slip rings, brushes, or external rotor resistance. Therefore, it offers:

1. Simple and rugged construction
2. Low maintenance requirement
3. High reliability for continuous operation
4. Lower cost and compact construction
5. Good efficiency for pumping applications
6. No brushes or slip rings, reducing mechanical wear
7. Suitable operation with commonly available three-phase power supplies

For a water pump, the load generally requires reliable and continuous rotational operation rather than extremely high starting torque. Therefore, the squirrel-cage induction motor provides a practical and economical solution.

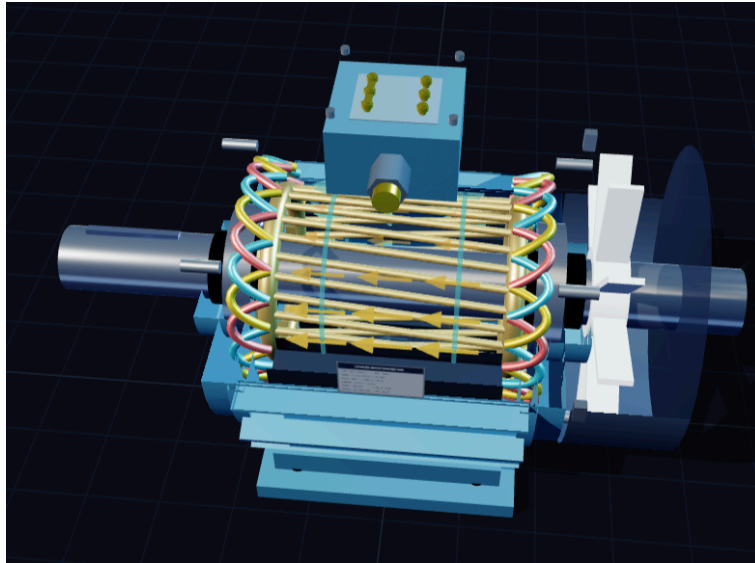


Fig A: Cutaway View of Squirrel Cage Motor

1.3 Motor Reconstruction

The reconstruction process involves restoring the motor to a functional operating condition. The major steps include:

Motor reconstruction and inspection

Stator teardown and preparation

Copper winding/rewinding of the stator

Proper arrangement and reconnection of the three-phase windings

Retention and inspection of the squirrel-cage rotor

The reconstruction process is carried out while maintaining the required electrical and mechanical parameters of the motor.

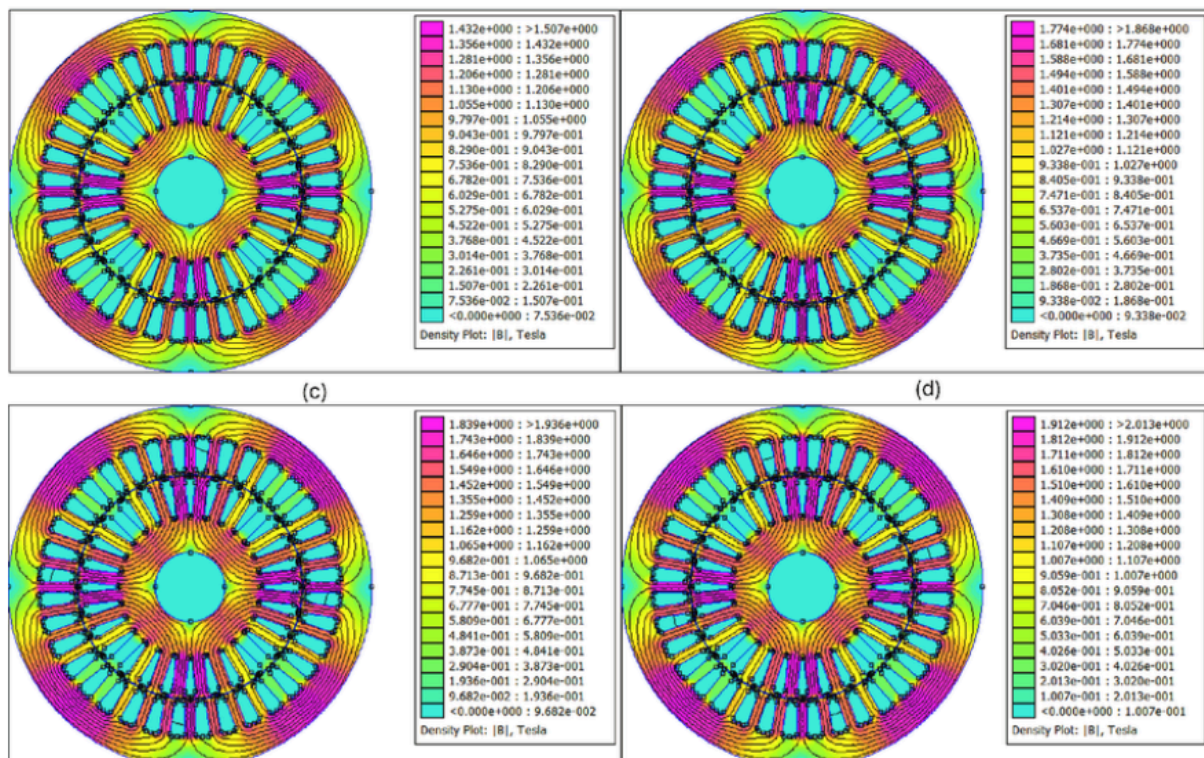


Fig B: Construction & Magnetic Flux

1.4 Parameter Extraction and MATLAB Modeling

After reconstruction, the necessary electrical and mechanical parameters of the motor are identified. These parameters are used to develop a mathematical model of the motor in MATLAB. The torque-speed characteristics are then analyzed to evaluate the motor's operating behavior under different speed and load conditions. This modeling provides an effective way to understand the performance of the reconstructed induction motor before its practical application.

1.5 Y-Connection and Electrical Configuration

The reconstructed motor is configured in a Y (star) connection for the three-phase supply. The applied line voltage is:

with a supply frequency of:

For a balanced star connection, the phase voltage is:

Thus, each stator phase receives approximately 230.94 V.

1.6 Shaft Coupling and Water Pumping Application

The final stage of the project involves mechanically coupling the motor shaft to the water-pumping mechanism. The rotational mechanical output of the induction motor is transferred through the shaft coupling to drive the pump. The performance of the system can then be evaluated in terms of the resulting water flow rate. Hence, the project connects the electrical reconstruction and theoretical analysis of the motor with a practical mechanical application.

1.7 Project Roadmap

The overall project methodology can therefore be summarized as:

Motor Reconstruction → Stator Teardown → Copper Rewinding → Parameter Extraction → MATLAB Torque-Speed Modeling → Shaft Coupling → Flow Rate Evaluation

This systematic approach allows the reconstructed squirrel-cage induction motor to be electrically analyzed, experimentally validated, and finally implemented for a practical water pumping application

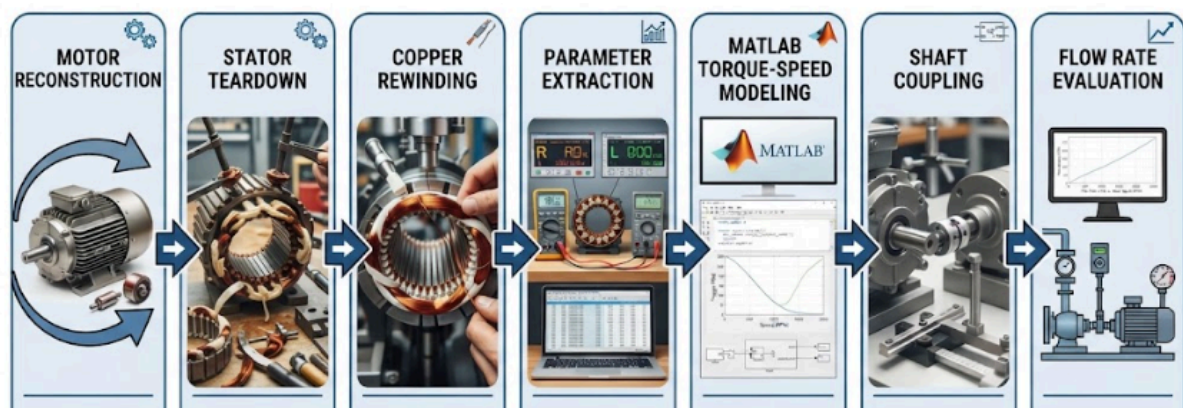


Fig C: Project Steps

Chapter 2: The Induction Motor Characterization

2.1 Winding Details

Parameter	Value
Rated power	.5 hp
Rated voltage	400V
Rated current	1.3 A
Frequency	50Hz
Rated speed	1400 rpm
Stator connection	Wye
Number of poles	4
Number of slots	24

Table 1: Motor Details

Basic Winding Calculations

The synchronous speed of the stator field is

$$N_s = 120f/P = (120 \times 50)/4 = 1500rpm$$

At the rated speed of 1400 rpm, the slip is

$$s = (N_s - N_r) / N_s = (1500 - 1400) / 1500 = 0.0667$$

The 24 slots are shared among 4 poles and 3 phases, so:

- Slots per pole = $24 / 4 = 6$
- Slots per pole per phase, $q = 24/(4 \times 3) = 2$
- Electrical angle between adjacent slots,
 $\gamma = (P/2 \times 360^\circ)/24 = 30^\circ$

Each phase therefore occupies 2 adjacent slots under every pole (a phase belt of $2 \times 30^\circ = 60^\circ$). The three phase belts are displaced by 120 electrical degrees (4 slots) from each other. This displacement is what allows three currents that are 120° apart in time to produce a single rotating magnetic field.

Basic Winding Arrangement

The winding is a single-layer, three-phase, star-connected winding with the following data:

- Total number of coils = 12 (one coil side per slot, 24 coil sides in total)
- Coils per phase = 4, connected in series
- Coil groups per phase = 2 (one per pole pair), each group holding $q = 2$ coils
- Turns per coil = 99
- Turns per phase = $4 \times 99 = 396$
- Conductors per slot = 99
- Total conductors = $24 \times 99 = 2376$
- Coil span: the two sides of each coil are separated by a gap of 4 slots

The four coils of each phase are joined in series, so that the current flows in opposite directions under adjacent poles. The three phases are then star-connected, with the three phase ends joined at the neutral point. The three free ends are brought out to terminals. In star connection the phase voltage and phase current are

About Y Connection

As $V_{\text{phase}} = V_{\text{line}}/\sqrt{3}$ and $I_{\text{phase}} = I_{\text{line}}$

Phase	Coil 1	Coil 2	Coil 3	Coil 4
A	1-6	7-12	13-18	19-24
B	5-10	11-16	17-22	23-4
C	9-14	15-20	21-2	3-8

Table 2: 3 Phase ReWinding Details

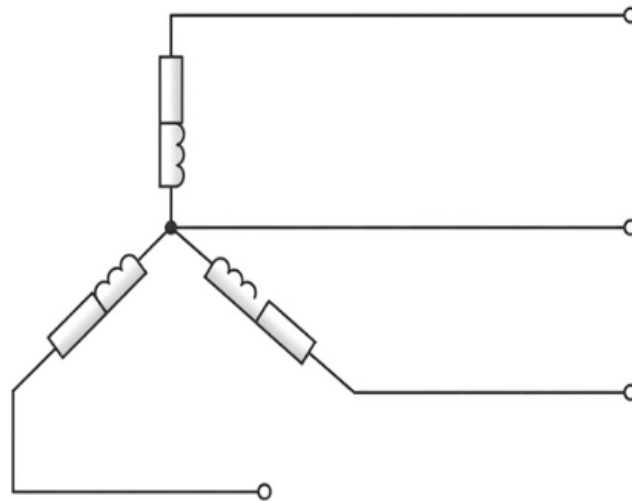


Fig 1: Y Phase Connection

2.2 The Middle Ground: Why a 4-Pole Motor?

Every induction motor design begins with one basic decision: how many magnetic poles the stator should have. This choice fixes the speed of the machine, and speed decides how much torque is left to drive the pump. A 4-pole configuration was selected for this project because it is the middle ground between speed and torque, giving the pump enough flow and enough pressure at the same time.

Poles and Synchronous Speed:

When three-phase currents flow through the stator winding, they create a magnetic field that rotates around the air gap, and the rotor follows it. The speed of this rotating field, the synchronous speed, depends only on the supply

frequency and the number of poles: more poles make the field turn slower, fewer poles make it turn faster.

$$N_s = 120 f / P$$

Here N_s is the synchronous speed in rpm, f is the supply frequency in Hz and P is the number of poles. On the 50 Hz supply, a 4-pole stator gives $N_s = (120 \times 50) / 4 = 1500$ rpm. The rotor always runs slightly slower than the field (slip), so with a typical full-load slip of 3–5 % the expected rotor speed is about 1425–1455 rpm.

The Speed–Torque Trade-off

Mechanical power is torque multiplied by speed, so for the same power a slower motor delivers more torque and a faster motor delivers less. Choosing the number of poles is therefore like choosing a gear: a 2-pole motor is high gear (fast, low torque), an 8-pole motor is low gear (strong, but slow) and a 4-pole motor is the middle gear. The options at 50 Hz are compared below, with torque worked out per kilowatt of output ($T = P / \omega$) at synchronous speed.

- **2-pole (3000 rpm, ≈ 3.2 N·m per kW):** fast but low in torque; noisier, with more bearing wear, so harsh on the pump.
- **4-pole (1500 rpm, ≈ 6.4 N·m per kW), selected:** balanced and smooth, moderate in size; the best compromise.
- **6-pole (1000 rpm, ≈ 9.5 N·m per kW):** more torque but slower, in a larger and costlier frame; flow is too low.
- **8-pole (750 rpm, ≈ 12.7 N·m per kW):** very high torque but very slow and bulky; too slow for useful flow.

Why 4 Poles Suit Water Pumping

Balanced flow and pressure: about 1500 rpm moves a useful volume of water, while the torque available at this speed gives excellent, steady hydraulic pressure.

Strong starting and running torque: it can drive both candidate pumps, the centrifugal and the regenerative type, the latter placing high starting-torque and pressure demands on the motor.

Perfect and durable: a moderate speed keeps vibration, noise and bearing wear low for long agricultural and domestic duty.

Easy to build: 24 slots divide evenly among 4 poles and 3 phases (2 slots per pole per phase), so the winding is symmetrical and practical to wind by hand.

2.3 Three-Phase Construction: The Stator Winding

The stator is the heart of the fabrication work: a 24-slot laminated core carrying a 4-pole, three-phase winding. The winding data are summarised below.

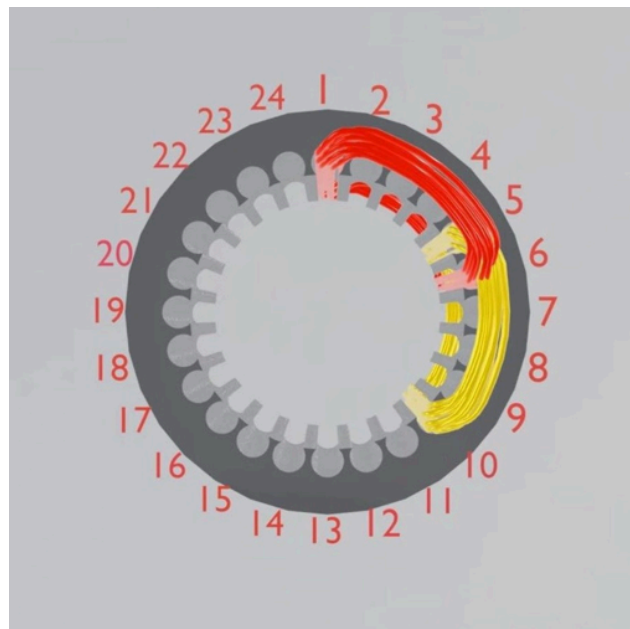


Fig 2: How 3 Phase Builds

Winding Design Data

- **Stator slots:** 24
- **Poles and phases:** 4 poles, 3 phases
- **Conductor:** 24 SWG super-enamelled copper wire
- **Turns:** 99 per slot
- **Coils:** 12 in total (4 per phase), placed two slots apart
- **Vacant slots:** 4

- **Terminal leads:** 24 (8 per phase)
- **Winding type:** single running winding, no starting coil
- **Insulation:** cotton-type MPR fibre tube and varnish

Slot Distribution

The slot arrangement follows directly from the slot, pole and phase counts:

- Slots per pole = $24 / 4 = 6$
- Slots per pole per phase, $q = 24 / (4 \times 3) = 2$
- Electrical angle per slot = $(2 \times 360^\circ) / 24 = 30^\circ$
- Phase displacement = $120^\circ / 30^\circ = 4$ slots

Each phase therefore starts four slots after the previous one, giving the exact 120° electrical shift needed for a balanced rotating field.

Coil Placement and the Two-Slot Gap

The winding is laid coil by coil. After one coil is inserted, the next is placed two slots further along. This gap is not arbitrary: it equals the number of slots per pole per phase ($q = 2$), which depends on the pole count and therefore on the rated speed, so a motor of different speed would need a different gap. Four slots are left vacant, appearing as the four gaps between coil groups in the winding layout, one per pole. In all, 12 coils of 99 turns are used: four coils ($4 \times 99 = 396$ turns) per phase and 1188 turns in the stator.

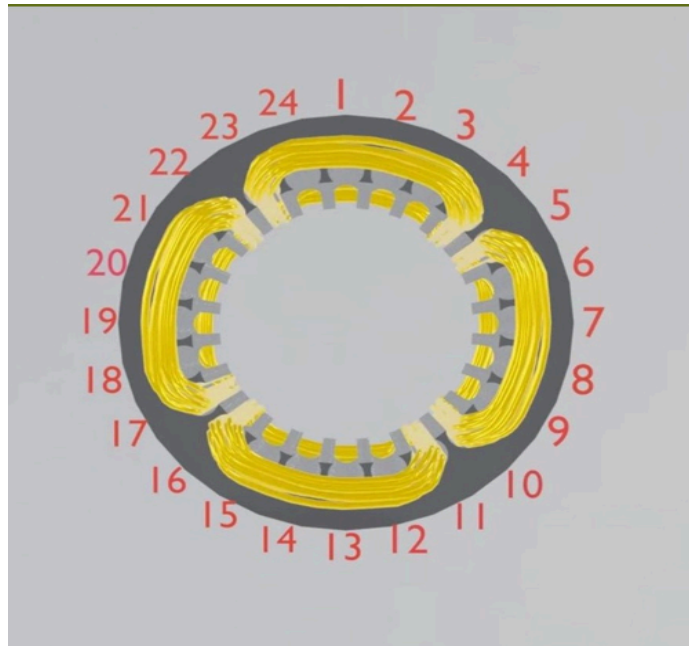


Fig 3: Single Phase Winding

Terminal Leads and Phase Formation

Each coil has two ends, so 12 coils bring out 24 terminal leads. These are divided into three groups of eight, one group per phase, each holding the ends of that phase's four coils. Bringing out every lead keeps the connections flexible: coils can be joined in series or parallel and the phases in star or delta.

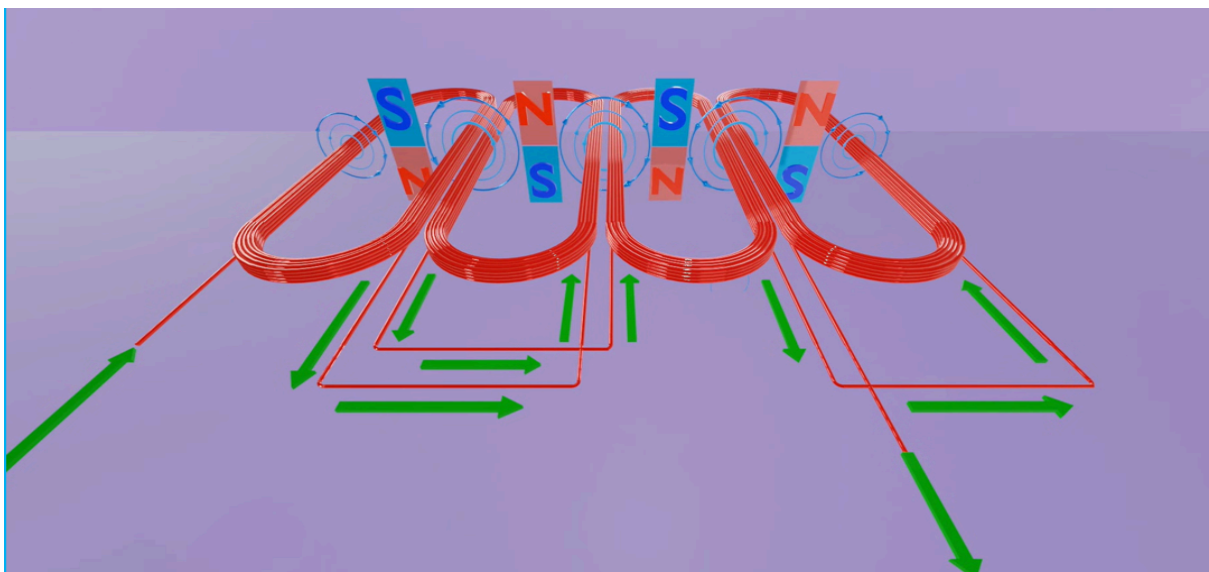


Fig 4: Current Flow Direction of 3 Phase Winding

Running Winding Only

A single-phase motor needs a separate starting coil because its field only pulses instead of rotating. A three-phase supply creates a rotating field on its own, so this motor is self-starting and carries a single set of coils, the running winding, with no starting winding or capacitor.

Chapter 3: Construction of The Induction Motor

3.1 Stator Details

The stator is the stationary part of the three-phase induction motor and contains the winding system responsible for producing the rotating magnetic field. In the reconstructed motor, the stator was examined in detail to determine its physical dimensions and winding arrangement.

The measured inner diameter of the stator is 6.5 cm, while the outer diameter is 13 cm. The stator slot length was measured as 53 mm. These dimensions provide the physical basis for arranging the three-phase winding inside the stator core.

The stator contains a total of 24 slots. Since the motor is a three-phase machine, the slots are distributed among the three phases. The winding arrangement therefore provides 8 slots per phase. This arrangement allows the three-phase winding to generate the required rotating magnetic field when supplied by a balanced three-phase voltage.

The distribution of the winding among the 24 slots is an important part of the reconstruction process. Proper placement of the coil sides is necessary to obtain the desired phase displacement and electromagnetic operation of the machine.



Fig A: Stator Look

3.2 Stator Winding Arrangement

The reconstructed stator uses a three-phase winding configuration. The winding is arranged so that the three phases are distributed throughout the available stator slots.

The project uses a total of **24 stator slots**, with **8 slots allocated to each phase**. The winding arrangement is designed to maintain the required electrical displacement between the three phases. This arrangement allows the motor to develop a rotating magnetic field when connected to a three-phase supply.

The coil arrangement and phase distribution are important because the generated magnetic field determines the electromagnetic torque and rotational behavior of the motor. Therefore, careful reconstruction of the winding layout is necessary for obtaining the desired motor performance.

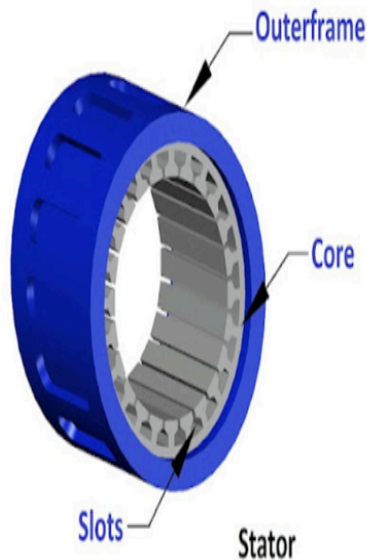


Fig B: Stator Frame

3.3 Stator Binding and Protection

After completing the winding arrangement, the stator winding requires mechanical and environmental protection. In the reconstructed motor, **cotton cloth and resin** are used to bind and secure the winding.

The binding arrangement helps prevent unwanted movement and vibration of the winding during motor operation. It also improves the mechanical stability of the coils inside the stator slots.

The use of resin additionally improves the insulation and provides protection against environmental effects. The winding arrangement therefore provides several benefits, including:

- Prevention of wire vibration
- Improved electrical insulation
- Better heat transfer
- Protection against moisture and dust

These features are important for maintaining the reliability and durability of the reconstructed induction motor.

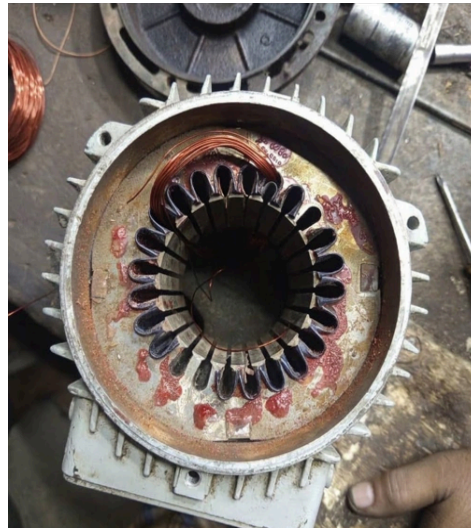


Fig C: Rewinding

3.4 Rotor Details

The rotor is the rotating component of the induction motor and interacts electromagnetically with the rotating magnetic field produced by the stator. The rotor construction influences the starting characteristics, efficiency, speed regulation, and maintenance requirements of the motor.

According to the project design, the motor construction provides the advantages of reliable operation and relatively low maintenance. The absence of sparking is also an important operational advantage. The motor is capable of providing high efficiency and relatively constant speed during normal operation.

However, the rotor arrangement also has certain limitations. The project identifies **low starting torque** and **difficulty in speed control** as disadvantages.

In addition, the rotor resistance cannot be changed externally in the described configuration. These characteristics influence the suitability of the motor for different applications.

Overall, the construction of the stator and rotor was an important part of the motor reconstruction process. The measured stator dimensions, 24-slot three-phase winding arrangement, insulation and binding system, and rotor characteristics provide the structural foundation for the subsequent electrical testing and parameter characterization of the machine.



Fig D: Rotor Construction



Fig E: Squirrel Cage Rotor

3.5 Insulation Details

Proper insulation is essential for reliable operation of the stator winding. In the reconstructed motor, a **three-phase Y-connected winding** is used. The winding is provided with **MPR tube cotton-type insulation**, and the insulation is further treated through burnishing.

The purpose of the insulation system is to electrically isolate the winding conductors from one another and from the stator core. Proper insulation also helps protect the winding against mechanical movement, moisture, and other environmental effects.

The insulation arrangement is therefore an important part of the reconstruction process because the electrical performance and safety of the motor depend on maintaining adequate insulation between the winding and other conductive parts.

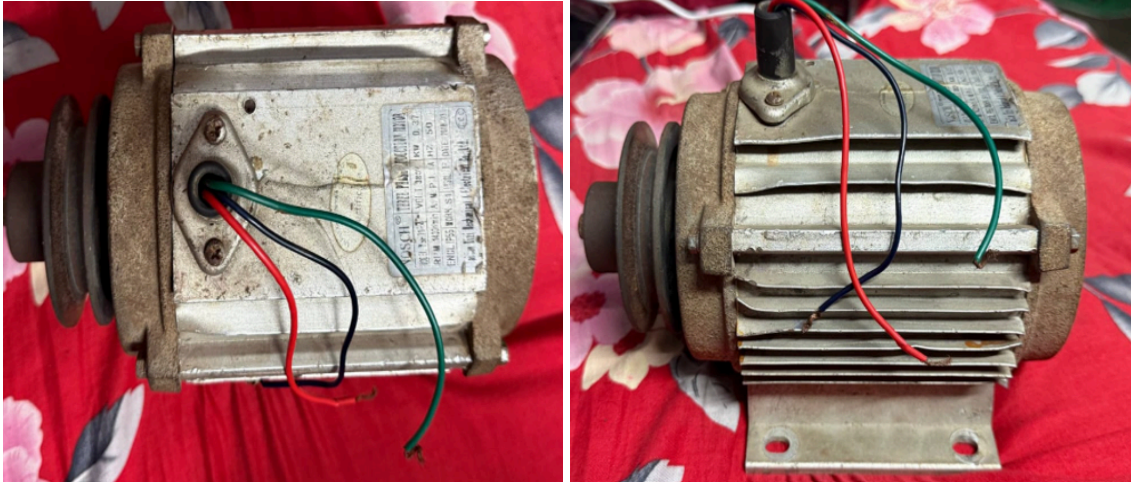


Fig F: Final Induction Motor



Fig G: Motor Rating Part

Chapter 4: Parameter Characterization

4.1 Testing Phase

After completing the reconstruction of the induction motor, electrical tests were carried out to determine the parameters required for analyzing its equivalent circuit and performance.

Three principal tests were considered:

1. **DC bridge measurement**
2. **No-load test**
3. **Blocked-rotor test**

These tests provide the electrical quantities required to determine the stator resistance, rotor-related parameters, core-loss resistance, and leakage and magnetizing reactances.

The obtained parameters were subsequently used to construct the equivalent circuit of the induction motor and to perform further analysis and simulation.

The Value We Got From Testing-

Name	DC Test	No Load Test	Blocked Rotor Test
Current(I)	1A	1.15A	1A
Voltage(V)	56.2V	380V	100V
Power(P)	-	185W	139W

Table 2: Testing Values

4.2 DC Test

The DC test was performed to determine the stator winding resistance of the three-phase induction motor.

The measured values were:

$$V_{dc} = 56.2 \text{ V}$$

$$I_{dc} = 1 \text{ A}$$

For the star-connected stator winding, the measured resistance corresponds to two phase resistances in series. Therefore,

$$2R_1 = V_{dc} \div I_{dc}$$

Substituting the measured values,

$$2R_1 = 56.2 \div 1$$

$$2R_1 = 56.2 \text{ } \Omega$$

Therefore,

$$R_1 = 56.2 \div 2$$

$$R_1 = 28.1 \text{ } \Omega$$

Hence, the stator winding resistance is

$$\mathbf{R_1 = 28.1 \text{ } \Omega}$$

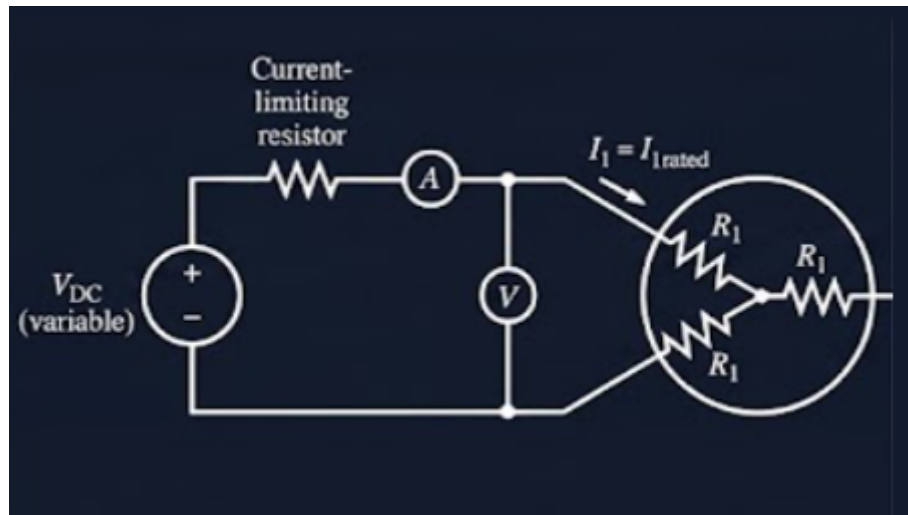


Fig 1: DC Test Circuit

4.3 No-Load Test Calculation

The no-load test was performed to determine the magnetizing branch parameters and the losses of the induction motor.

The measured values were

$$V_{\text{line}} = 380 \text{ V}$$

$$I_{\text{line}} = 1.15 \text{ A}$$

For the Y-connected winding, the phase voltage is

$$V_{\text{phase}} = V_{\text{line}} \div \sqrt{3}$$

$$V_{\text{phase}} = 380 \div \sqrt{3}$$

$$V_{\text{phase}} = 219.39 \text{ V}$$

Since the line current is equal to the phase current in a Y connection,

$$I_{\text{phase}} = I_{\text{line}}$$

$$I_{\text{phase}} = 1.15 \text{ A}$$

No-Load Impedance

The no-load impedance is calculated from the phase voltage and phase current.

$$Z = V_{\text{phase}} \div I_{\text{phase}}$$

$$Z = 219.39 \div 1.15$$

$$Z = 190.77 \, \Omega$$

Therefore,

$$\mathbf{Z = 190.77 \, \Omega}$$

At no load, the impedance is mainly associated with the stator leakage reactance and magnetizing reactance. Hence,

$$Z \approx X_1 + X_m$$

Therefore,

$$\mathbf{X_1 + X_m = 190.77 \, \Omega}$$

Stator Copper Loss

The stator copper loss during the no-load test is calculated as

$$P_{\text{stator}} = 3 I_{\text{phase}}^2 R_1$$

Substituting the measured values,

$$P_{\text{stator}} = 3 \times 1.15^2 \times 28.1$$

$$P_{\text{stator}} = 3 \times 1.3225 \times 28.1$$

$$P_{\text{stator}} = 111.49 \, \text{W}$$

Therefore,

$$\mathbf{P_{\text{stator}} \approx 111.486 \, \text{W}}$$

Rotational and Core-Related Loss

The project data gives the remaining loss component as

$$P_{rot} = 73.514 \text{ W}$$

Thus,

$$\mathbf{P_{rot} = 73.514 \text{ W}}$$

This value is used in determining the shunt core-loss resistance.

Core-Loss Resistance

The core-loss resistance is calculated using the phase voltage and the corresponding loss.

$$R_c = V_{phase}^2 \div P_{rot}$$

Substituting the values,

$$R_c = 219.39^2 \div 73.514$$

$$R_c = 48132.10 \div 73.514$$

$$R_c \approx 654.75 \text{ } \Omega$$

Therefore,

$$\mathbf{R_c = 654.754 \text{ } \Omega}$$

Hence, the core-loss resistance obtained from the no-load test is approximately **654.754 Ω** .

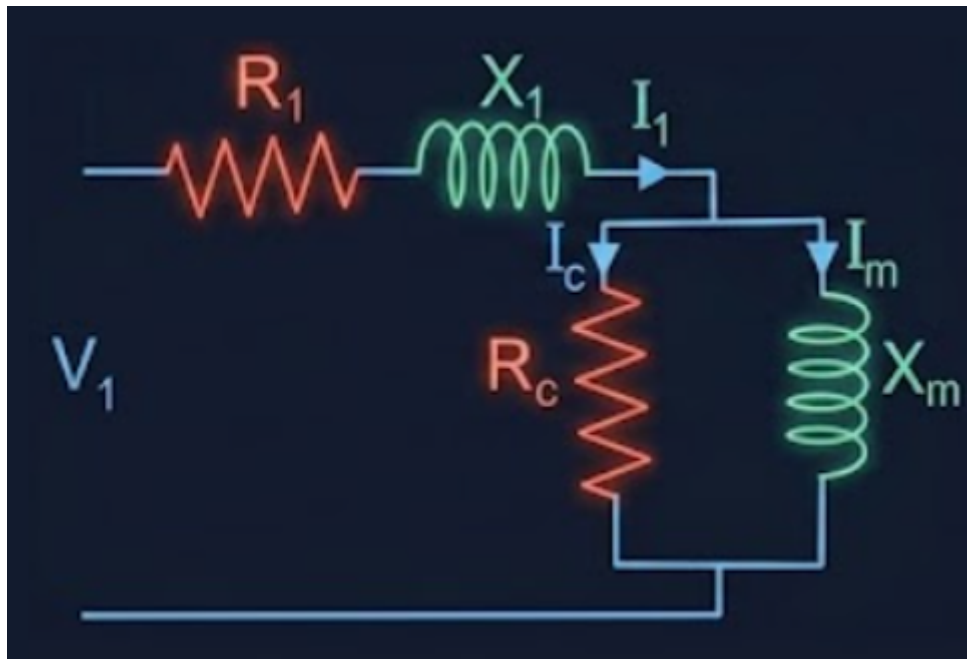


Fig 2: No Load Test Circuit

4.4 Blocked-Rotor Test Calculation

The blocked-rotor test was performed to determine the equivalent series resistance and leakage reactance of the induction motor.

During the test, the rotor was prevented from rotating.

The measured values were

$$V_{\text{line}} = 100 \text{ V}$$

$$I_{\text{line}} = 1 \text{ A}$$

$$P = 139 \text{ W}$$

The rotor was tested at a frequency of

$$f = 15 \text{ Hz}$$

Blocked-Rotor Phase Voltage

Since the motor winding is Y-connected,

$$V_{\text{phase}} = V_{\text{line}} \div \sqrt{3}$$

$$V_{\text{phase}} = 100 \div \sqrt{3}$$

$$V_{\text{phase}} = 57.73 \text{ V}$$

The phase current is

$$I_{\text{phase}} = I_{\text{line}}$$

$$I_{\text{phase}} = 1 \text{ A}$$

Blocked-Rotor Impedance

The blocked-rotor impedance is

$$Z_{\text{br}} = V_{\text{phase}} \div I_{\text{phase}}$$

$$Z_{\text{br}} = 57.73 \div 1$$

$$Z_{\text{br}} = 57.73 \Omega$$

Therefore,

$$\mathbf{Z_{br} = 57.73 \Omega}$$

Impedance Angle

The power factor during the blocked-rotor test is

$$\cos \phi = P \div 3 V_{\text{phase}} I_{\text{phase}}$$

Substituting the measured values,

$$\cos \phi = 139 \div 3 \times 57.73 \times 1$$

$$\cos \phi = 139 \div 173.19$$

$$\cos \varphi = 0.8024$$

Therefore,

$$\varphi = \cos^{-1} 0.8024$$

$$\varphi \approx 36.62^\circ$$

Hence,

$$\varphi = \mathbf{36.62^\circ}$$

Combined Resistance

The total resistance of the stator and rotor referred to the stator side is

$$R_1 + R_2 = Z_{br} \cos \varphi$$

$$R_1 + R_2 = 57.73 \times \cos 36.62^\circ$$

$$R_1 + R_2 \approx 46.33 \, \Omega$$

Therefore,

$$\mathbf{R_1 + R_2 = 46.33 \, \Omega}$$

Since the stator resistance obtained from the DC test is

$$R_1 = 28.1 \, \Omega$$

the rotor resistance referred to the stator is

$$R_2 = 46.33 - 28.1$$

$$R_2 = 18.23 \, \Omega$$

Therefore,

$$\mathbf{R_2 = 18.23 \, \Omega}$$

Total Leakage Reactance at 15 Hz

The total leakage reactance is

$$X' = Z_{br} \sin \phi$$

$$X' = 57.73 \times \sin 36.62^\circ$$

$$X' \approx 34.43 \, \Omega$$

Therefore,

$$\mathbf{X' = 34.43 \, \Omega \text{ at } 15 \text{ Hz}}$$

Leakage Reactance at 60 Hz

Leakage reactance is proportional to frequency. Therefore,

$$X_{60} = X_{15} \times 60 \div 15$$

$$X_{60} = 34.43 \times 60 \div 15$$

$$X_{60} = 137.72 \, \Omega$$

Approximately,

$$\mathbf{X = 137.74 \, \Omega \text{ at } 60 \text{ Hz}}$$

Stator and Rotor Leakage Reactance

Assuming that the total leakage reactance is equally divided between the stator and rotor,

$$X_1 = X_2 = X \div 2$$

$$X_1 = X_2 = 137.74 \div 2$$

$$X_1 = X_2 = 68.87 \, \Omega$$

Therefore,

$$\mathbf{X_1 = 68.87 \, \Omega}$$

$$\mathbf{X_2 = 68.87 \, \Omega}$$

Magnetizing Reactance

From the no-load and blocked-rotor test analysis, the magnetizing reactance obtained for the equivalent circuit is

$$\mathbf{X_m = 121.9 \, \Omega}$$

Thus, the final equivalent-circuit parameters obtained from the electrical tests are

$$R_1 = 28.1 \, \Omega$$

$$R_2 = 18.23 \, \Omega$$

$$R_c = 654.754 \, \Omega$$

$$X_1 = 68.87 \, \Omega$$

$$X_2 = 68.87 \, \Omega$$

$$X_m = 121.9 \, \Omega$$

These parameters are used to represent the electrical behavior of the reconstructed three-phase induction motor and form the basis for further torque-speed and performance analysis.

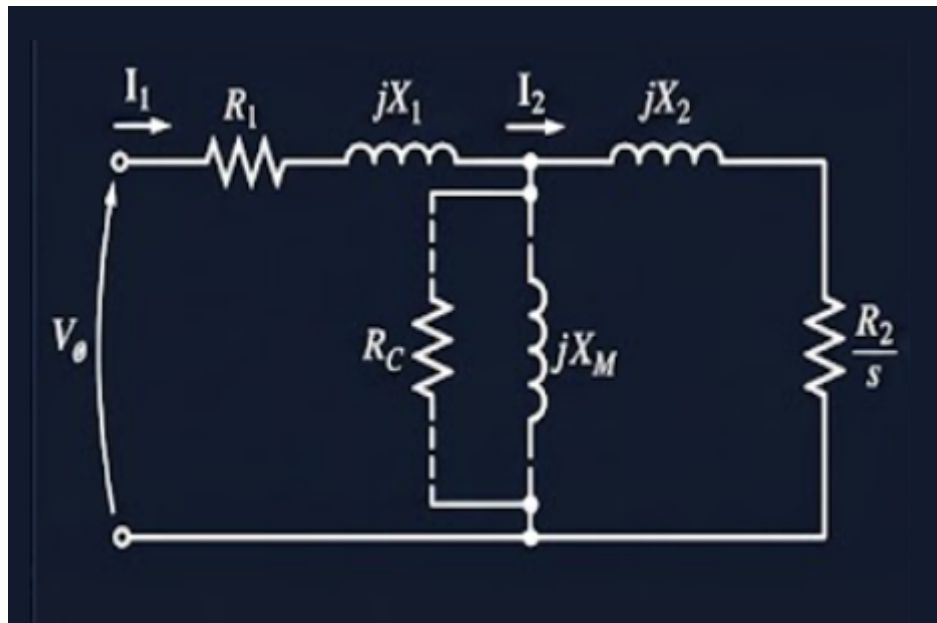


Fig 3: Blocked Rotor Test Circuit

4.5 Equivalent Circuit Parameters

The electrical tests provided the parameters required for representing the reconstructed induction motor by an equivalent circuit.

The final equivalent-circuit parameters obtained from the project are:

Parameter	Value(Ω)
R1	28.1
R2	18.23
Rc	654.754
X1	68.87

X2	68.87
Xm	121.9

Table 2: Parameter Values

These parameters provide a mathematical representation of the electrical behavior of the reconstructed motor. The stator resistance and leakage reactance represent the stator-side impedance, while the rotor resistance and leakage reactance represent the rotor-related effects referred to the stator side. The core-loss resistance and magnetizing reactance represent the shunt branch associated with the magnetic behavior of the motor.

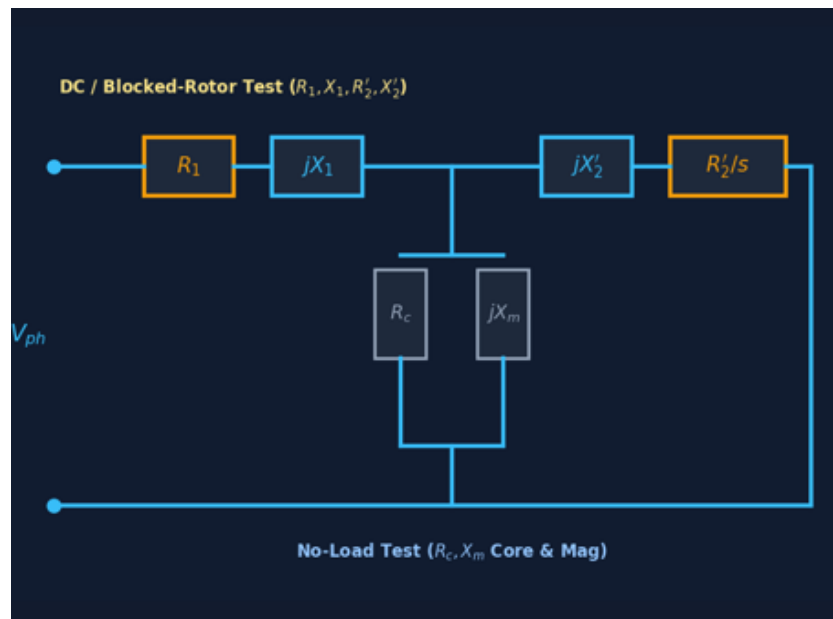


Fig 4: Induction Motor Equivalent Circuit

4.6 Motor Design Class

Based on the project characterization, the motor is identified with **Design Class A** characteristics. The design is associated with **high running efficiency** and **stable speed**.

The classification provides an indication of the expected operating characteristics of the motor. The obtained electrical parameters can subsequently be used to analyze the torque-speed behavior and other operating characteristics of the machine.

4.7 Summary of Electrical Characterization

The reconstruction and electrical characterization of the induction motor involved determining both the physical and electrical properties of the machine. The DC test established the stator resistance, while the no-load test provided information about the magnetizing branch and losses. The blocked-rotor test was then used to determine the combined resistance and leakage reactance parameters.

The final equivalent-circuit parameters obtained were: These experimentally obtained values form the basis for further analysis and simulation of the reconstructed three-phase induction motor. They allow the electrical behavior of the machine to be represented through an equivalent circuit and provide the necessary parameters for subsequent performance evaluation.

Chapter 5: MATLAB Simulation

5.1 Thevenin Calculation

The Thevenin equivalent circuit simplifies the complex motor circuit into a single voltage source and impedance connected to the rotor branch:

- Per-Phase Voltage (V_{ph}): Calculated from the line voltage ($V_{line} = 400 \text{ V}$) using the star connection formula:

$$V_{ph} = \frac{V_{line}}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230.94 \text{ V}$$

- Magnetizing Impedance (Z_m): Represents the core loss resistance (R_c) and magnetizing reactance (X_m) in parallel:

$$Z_m = \frac{R_c \cdot jX_m}{R_c + jX_m} = 22.38 + j120.21 \Omega$$

- Stator Impedance (Z_1): Combines the stator resistance (R_1) and stator leakage reactance (X_1):

$$Z_1 = R_1 + jX_1 = 28.1 + j68.87 \Omega$$

- Thevenin Impedance (Z_{th}) & Voltage (V_{th}): Derived by combining the source and magnetizing branches to find equivalent parameters:

$$Z_{th} = \frac{Z_1 \cdot Z_m}{Z_1 + Z_m} = 14.05 + j43.93 \Omega \quad (R_{th} = 14.05 \Omega, X_{th} = 43.93 \Omega)$$

$$V_{th} = V_{ph} \cdot \left| \frac{Z_m}{Z_1 + Z_m} \right| = 143.2 \text{ V}$$

VTh	XTh	RTh
143.2	43.93	14.05

Table 1: Thevenin Parameter

5.2 Slip Calculations (Detailed Derivation & Explanation)

Slip (s) represents the relative speed difference between the synchronous magnetic field rotating speed (N_s) and the actual rotor mechanical speed (N), expressed as a fraction or percentage.

- **Formula:**

$$s = \frac{N_s - N}{N_s}$$

- **Why Slip Starts at 1.0 ($s_{start} = 1.0$):**

At the very beginning of motor startup, the rotor is completely stationary ($N = 0$ RPM). Substituting this into the slip equation:

$$s_{start} = \frac{N_s - 0}{N_s} = \frac{N_s}{N_s} = 1.0$$

This maximum slip value (**1.0** or **100%**) indicates that the relative speed between the stator field and rotor conductors is at its absolute maximum, inducing maximum rotor frequency and high starting current.

Max slip calculation: Slip at maximum torque (s_{max}) is the specific slip value where the induction motor develops its peak breakdown torque (T_{max}),

determined by the formula
$$s_{max} = \frac{R_2'}{\sqrt{R_{th}^2 + (X_{th} + X_2')^2}}$$
. Substituting the project's equivalent circuit parameters—specifically the referred rotor resistance $R_2' = 18.23 \Omega$, Thevenin resistance $R_{th} = 14.05 \Omega$, Thevenin reactance $X_{th} = 43.93 \Omega$, and referred rotor reactance $X_2' = 68.87 \Omega$ —we first sum the reactances to get **112.80 Ω** , square both components and add them inside the square root to yield approximately **113.67**. Finally, dividing the rotor resistance **18.23 Ω** by this denominator gives a slip value of $s_{max} \approx 0.16$ (or **16%**), which successfully establishes the breakdown torque speed at **1515 RPM**.

- **Why $s = 0.05$ (**5%**) was Chosen for Full Load:**

Under standard operating conditions, a well-designed 3-phase induction motor does not run at synchronous speed due to mechanical load resistance. A full-load slip of **0.05 (or 5%)** is universally selected as an industry-standard design baseline for standard motors. This ensures optimal balance between operating efficiency, slip-ring losses, and speed regulation under nominal mechanical load.

S(Start)	S(Max)	S(Rated)
1.00	0.16	0.05

Table 2: Slip Values

5.3 Torque Calculations (τ / T) with Formulas & Steps

Electromagnetic developed torque (T_{dev}) in a 3-phase induction motor is derived from the per-phase equivalent circuit parameters, taking into account the Thevenin equivalent voltage (V_{th}), impedance (R_{th}, X_{th}), referred rotor resistance (R_2'), referred rotor reactance (X_2'), and synchronous angular velocity (ω_{sync}).

- **General Developed Torque Formula:**

$$T_{dev} = \frac{3 \cdot V_{th}^2 \cdot \left(\frac{R_2'}{s}\right)}{\omega_{sync} \cdot \left[\left(R_{th} + \frac{R_2'}{s}\right)^2 + (X_{th} + X_2')^2 \right]}$$

1. Locked-Rotor / Starting Torque (T_{start}):

- At startup, slip $s = 1.0$. Substituting this into the general torque equation:

$$T_{start} = \frac{3 \cdot V_{th}^2 \cdot R_2'}{\omega_{sync} \cdot [(R_{th} + R_2')^2 + (X_{th} + X_2')^2]}$$

- **Calculation & Value:**

Given values: $V_{th} = 143.2 \text{ V}$, $R_{th} = 14.05 \Omega$, $X_{th} = 43.93 \Omega$, $R_2' = 18.23 \Omega$, $X_2' = 68.87 \Omega$, $\omega_{sync} = 188.5 \text{ rad/s}$.

$$T_{start} = \frac{3 \cdot (143.2)^2 \cdot 18.23}{188.5 \cdot [(14.05 + 18.23)^2 + (43.93 + 68.87)^2]} = 0.4 \text{ N} \cdot \text{m}$$

2. Maximum / Breakdown Torque (T_{max}):

- **Slip at Maximum Torque (s_{max}):** Breakdown torque occurs when the slip matches the critical resistance-to-reactance ratio:

$$s_{max} = \frac{R_2'}{\sqrt{R_{th}^2 + (X_{th} + X_2')^2}} = \frac{18.23}{\sqrt{(14.05)^2 + (43.93 + 68.87)^2}} \approx 0.16$$

- **Maximum Torque Formula & Value:**

$$T_{max} = \frac{3 \cdot V_{th}^2}{2 \cdot \omega_{sync} \cdot (R_{th} + \sqrt{R_{th}^2 + (X_{th} + X_2')^2})} = 1.3 \text{ N} \cdot \text{m}$$

Corresponding speed at breakdown:

$$N_{max} = (1 - s_{max}) \cdot N_s = (1 - 0.16) \cdot 1800 \approx 1515 \text{ RPM}$$

3. Full-Load Torque (T_{FL}):

- Evaluated at rated slip $s_{rated} = 0.05$:

$$T_{FL} = \frac{3 \cdot (143.2)^2 \cdot \left(\frac{18.23}{0.05}\right)}{188.5 \cdot \left[\left(14.05 + \frac{18.23}{0.05}\right)^2 + (43.93 + 68.87)^2\right]} = 0.8 \text{ N} \cdot \text{m}$$

Corresponding full-load speed: $N_{FL} = (1 - 0.05) \cdot 1800 = 1710 \text{ RPM}$

Position/Name	Torque(Nm)	Speed(rpm)
Start	0.4	0
Max	1.3	1515
Full Load/Rated	0.8	1710

Table 3: Torque Vs Speed

5.4 MATLAB Code Explanation and Structure

The MATLAB script begins with workspace clearing commands (`clc; close all;`) and defines core parameters like supply voltage (400 V), frequency (60 Hz), and pole count (4). It computes the Thevenin equivalents, builds a speed vector from 0 to synchronous speed ($N_s = 1800\text{ RPM}$) using `linspace()`, calculates electromagnetic torque across all slip values using the equivalent circuit formula, and extracts key operating points ($T_{start}, T_{max}, T_{FL}$) to display performance summaries in the command window.

5.5 App Interface Explanation

A custom-built MATLAB GUI application named **"Motor & Pump Dashboard"** serves as a centralized control hub. It features two interactive push buttons—**"View Torque-Speed Curve"** and **"View Pump Flow Rate Curve"**—allowing users to instantly visualize distinct electromechanical and hydrodynamic behaviors without interacting directly with raw code.

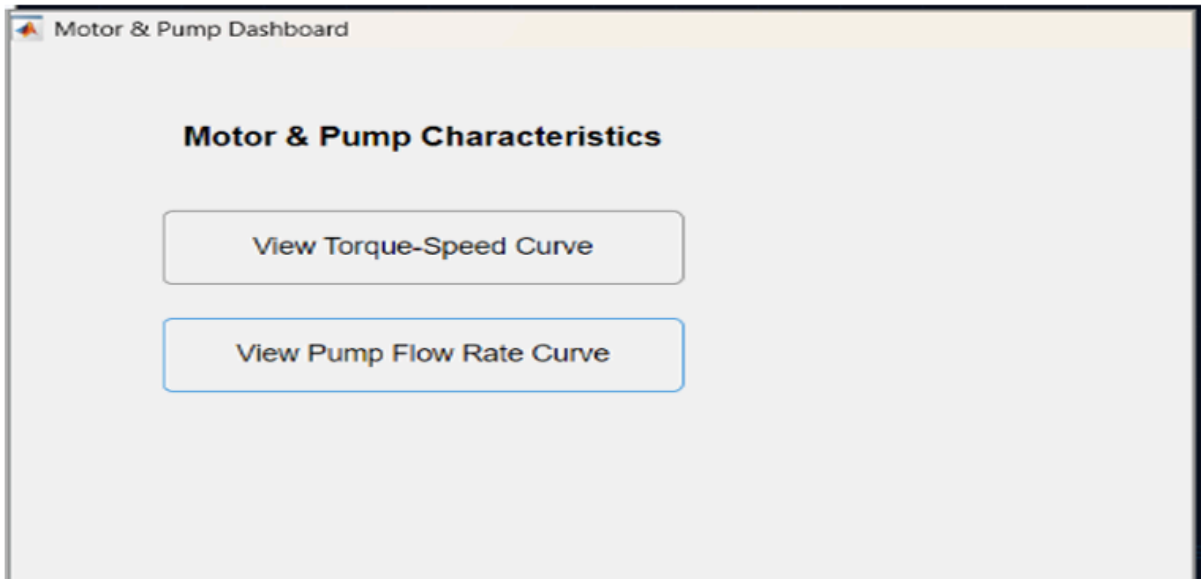


Fig1 :App Interface

5.6 Plotting The 2 Curves

1.Torque-Speed Characteristic Curve: Plots developed electromagnetic torque against rotor speed (N), highlighting critical points such as Locked-Rotor Torque ($0.4 \text{ N} \cdot \text{m}$), Pull-Up Torque, Breakdown Torque ($1.3 \text{ N} \cdot \text{m}$ at 1515 RPM), and Full-Load Operating Point ($0.8 \text{ N} \cdot \text{m}$ at 1710 RPM) alongside embedded machine specification data.

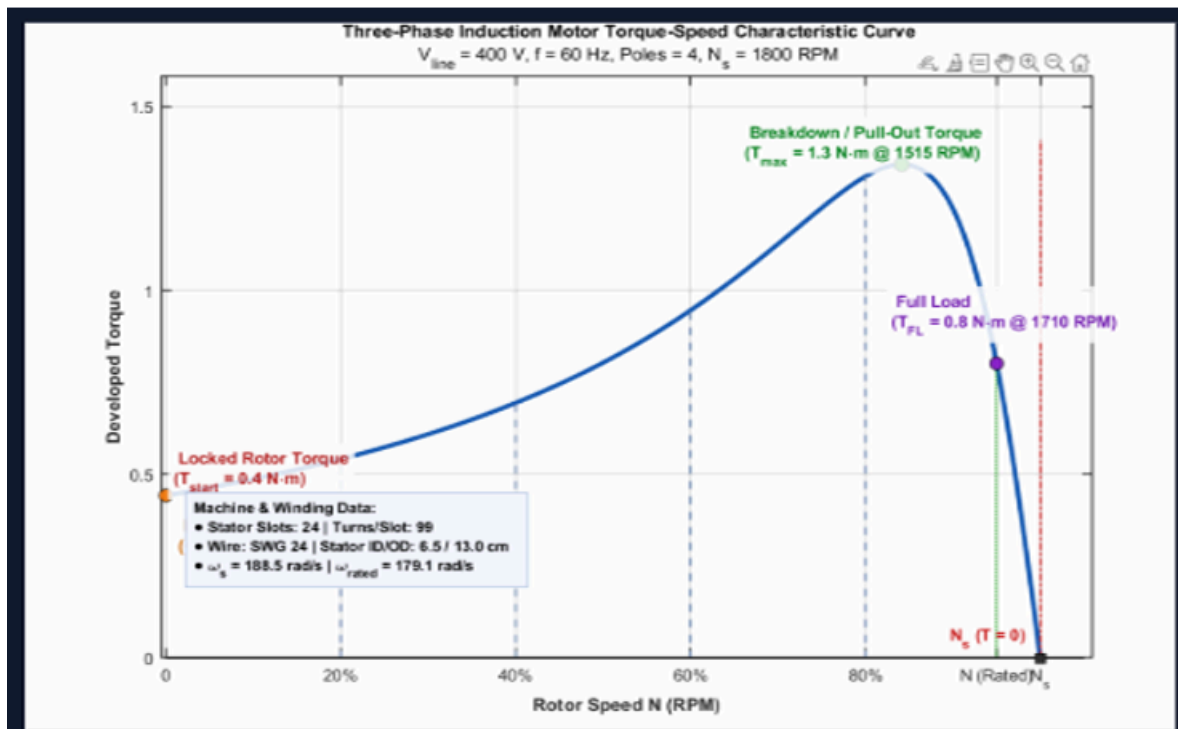


Fig 2 : Torque-speed Characteristic Curve

2. **Speed vs. Flow Rate Characteristic Curve:** Maps centrifugal pump mechanics where volumetric discharge flow rate (Q) is directly proportional to rotational shaft speed ($Q \propto \omega$).

- **Standstill** ($N = 0 \text{ RPM}$): $Q = 0.00 \text{ L/s}$ ($0.0 \text{ m}^3/\text{h}$).
- **Max Torque Stage** ($N \approx 1514.9 \text{ RPM}$): $Q = 10.63 \text{ L/s}$.
- **Rated Speed** ($N = 1710.0 \text{ RPM}$): $Q = 12.00 \text{ L/s}$ ($43.2 \text{ m}^3/\text{h}$).

Position/Name	Speed(rpm)	Flow Rate(L/s)
Start	0	0
Max	1515	10.63
Full Load/Rated	1710	12.00

Table 4: Speed Vs Flow Rate

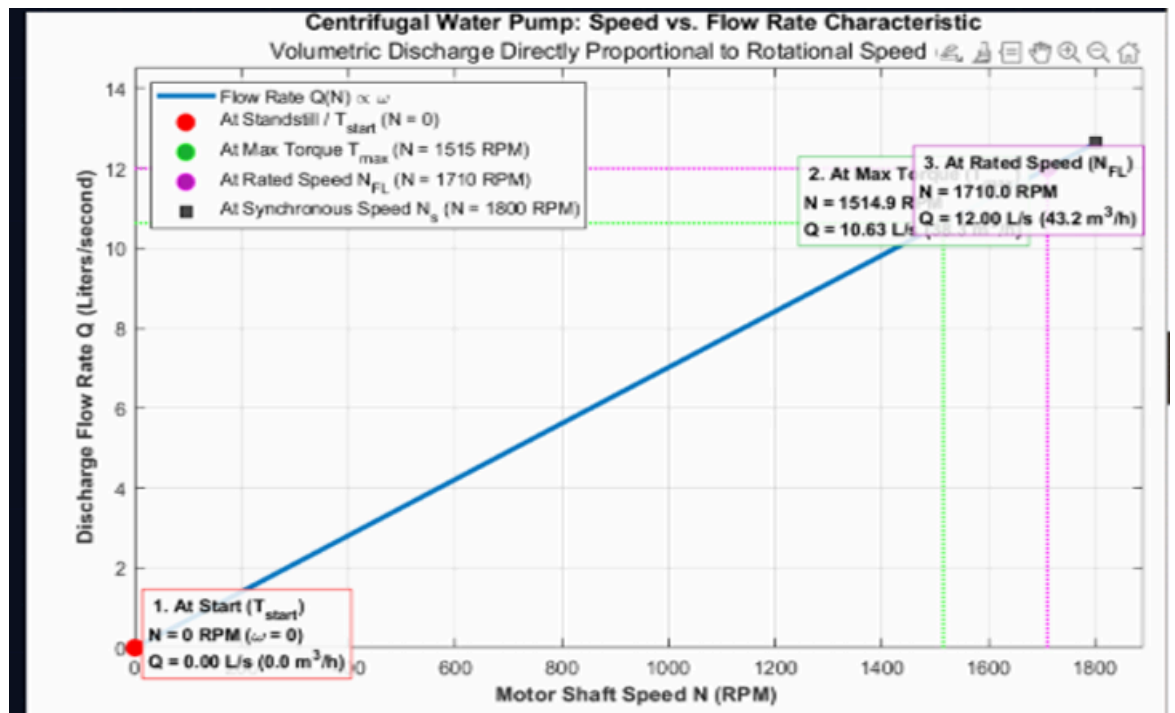


Fig 3: Speed vs Flow Rate Characteristic

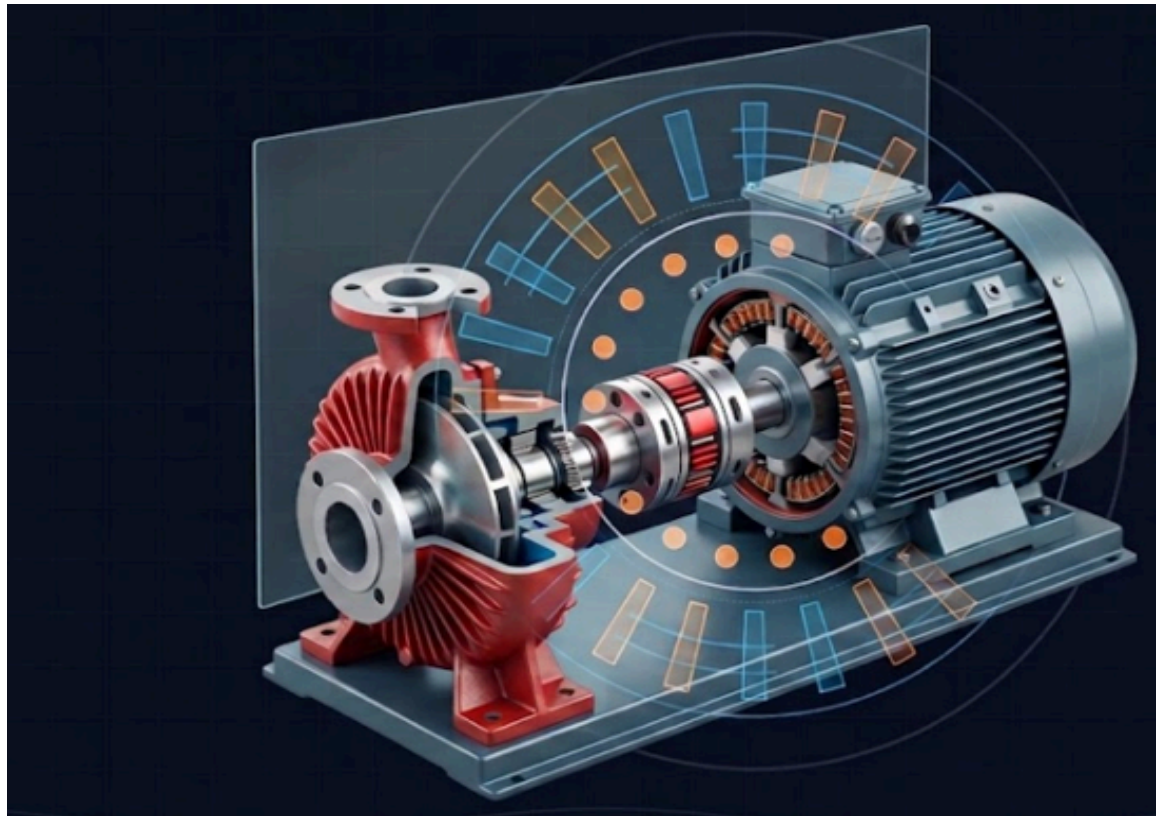


Fig 4: Induction Motor Coupling With Water Pump

Chapter 6: Limitations & Future Uses

6.1 Limitations

- **Limited Maximum Torque Capacity (T_{max}):** The motor has a limited breakdown torque capacity ($1.3 \text{ N} \cdot \text{m}$), restricting its ability to handle sudden, heavy mechanical overloads or severe transient load spikes without stalling.
- **High Inrush Current Stress:** Direct-on-line (DOL) starting draws excessive initial current, creating significant thermal degradation of the

stator coil insulation and momentary voltage drops across the local distribution transformer grid.

- **Hydraulic Flow Constraints:** Volumetric fluid delivery is limited at specific operating points, making single-stage configurations less optimal for high-volume or heavy-duty industrial and agricultural irrigation demands.
- **Speed Regulation Economic Trade-offs:** Maintaining smooth speed regulation and precise control under changing loads requires external hardware like Variable Frequency Drives (VFDs), which substantially increases overall project implementation costs.

6.2 Future Uses

- **Smart Farming:** Integrating the motor-pump system with digital microcontrollers, wireless IoT communication modules, and soil moisture sensors to automate irrigation networks, ensuring crops are watered dynamically based on real-time field dryness levels.
- **Green Energy Production:** Coupling the three-phase induction motor architecture with eco-friendly renewable energy sources, such as solar photovoltaic arrays or micro-wind turbines, to create sustainable, off-grid water pumping stations for remote agricultural regions.
- **Energy Storage Systems:** Utilizing advanced energy storage architectures to store surplus electrical or mechanical energy efficiently during low-demand periods, stabilizing power availability and improving overall system energy management.
- **Anomaly Detection:** Incorporating continuous sensor feedback loops to monitor winding temperatures and mechanical vibrations in real time, detecting abnormal overheating early and resolving heat-related issues to maximize equipment lifespan and reliability.

6.3 Conclusion

In conclusion, this project has successfully implemented and validated a comprehensive, end-to-end workflow for motor reconstruction and subsequent performance evaluation. By meticulously executing the seven-stage process—from initial assessment and physical refurbishment (teardown and copper rewinding) to analytical parameter extraction and MATLAB torque-speed modeling—the project has created a seamless bridge between physical repair and engineering analysis. The final evaluation, which integrated the reconstructed motor into a pump system to measure real-world flow rate, successfully validated the theoretical MATLAB model. This demonstrates a robust and effective methodology for not only restoring electric motor functionality but also for accurately predicting and verifying operational performance before deployment, ensuring system reliability and efficiency.

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[7] MathWorks, *Simscape Electrical: Modeling and Simulating Three-Phase Induction Motor Drives*, Natick, MA: The MathWorks, Inc., 2025.

Github

<https://github.com/anirudha175/Three-Phase-Induction-Motor-For-Water-Pumping-Applications#three-phase-induction-motor-for-water-pumping-applications>

Testing Video Link

https://drive.google.com/file/d/1P8U93nQypQxA6yjRkoNO3seB2DO7cxY4/view?usp=drive_link

Matlab Code

<https://github.com/anirudha175/Three-Phase-Induction-Motor-For-Water-Pumping-Applications/blob/main/MATLAB.M>

3D View

<https://github.com/anirudha175/Three-Phase-Induction-Motor-For-Water-Pumping-Applications/blob/main/3DVIEW.MP4>

Question & Answer

1. Q: What is the primary focus of the project regarding the induction motor?

A: The project focuses on the reconstruction, analysis, and practical application of a three-phase squirrel-cage induction motor specifically for water pumping systems.

2. Q: Why is a squirrel-cage induction motor preferred for water pumping applications?

A: It is chosen because of its simple and rugged construction, low maintenance requirements, high reliability for continuous operation, no brushes or slip rings to reduce mechanical wear, and cost-effectiveness.

3. Q: What is the selected pole configuration for the motor, and why was it chosen?

A: A 4-pole configuration was chosen because it serves as the ideal middle ground between speed and torque (providing $\sim 6.4 \text{ N}\cdot\text{m}$ per kW at synchronous speed), ensuring sufficient water flow and robust pressure while keeping vibration and bearing wear low.

4. Q: What are the physical dimensions and slot counts of the stator measured during reconstruction?

A: The stator has an inner diameter of 6.5 cm, an outer diameter of 13 cm, a slot length of 53 mm, and a total of 24 slots distributed among 3 phases (giving 8 slots per phase).

5. Q: How is the stator winding electrically connected to the three-phase power supply?

A: The stator winding is configured in a Y (star) connection, where each phase receives approximately 230.94 V from a 400 V line supply.

6. Q: What is the purpose of performing the DC test on the induction motor?

A: The DC test is performed to determine the stator winding resistance (R_1) of the three-phase induction motor, which was measured to be 28.1Ω .

7. Q: What parameters are determined using the no-load test?

A: The no-load test is used to determine the magnetizing branch parameters, rotational losses ($P_{rot} = 73.514W$), and the shunt core-loss resistance ($R_c \approx 654.75\Omega$).

8. Q: What is the purpose of the blocked-rotor test, and at what frequency was it performed?

A: The blocked-rotor test determines the equivalent series resistance and leakage reactance of the motor. It was performed at a reduced frequency of 15 Hz to accurately calculate parameters scaled to 60 Hz.

9. Q: What are the key torque performance values obtained from the MATLAB simulation?

A: The simulation yields a starting (locked-rotor) torque (T_{start}) of 0.4 Nm, a maximum breakdown torque (T_{max}) of 1.3 Nm at 1515 RPM, and a full-load torque (T_{FL}) of 0.8 Nm at 1710 RPM.

10. Q: What are some potential future enhancements proposed for this motor-pump system?

A: Future improvements include smart farming automation via IoT sensors, integration with renewable energy sources like solar photovoltaic arrays, energy storage implementation, and real-time thermal/vibration anomaly detection.

